

Algorithms: Spiking ConvNeXt with TTFS Analytic Mapping

Algorithm 1 Spiking ConvNeXt Block (TTFS Analytic Mapping with Min-Residual)

Input: $t_j \in \mathbb{R}^{N \times C \times H \times W}$ (input spike times), $t_{\min}$, $t_{\max}$
DWConv($\cdot$), $W1$, $W2$ (pointwise weights), D_{mid} , D_{out}
(learnable delays)

DropPath operator DropPath($\cdot$), force_positive $\in \{0,1\}$

Output: $t_{\text{out}} \in \mathbb{R}^{N \times C \times H \times W}$ (output spike times)

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1:  $x \leftarrow \text{DWConv}(t_j)$ 
2:  $(N, C, H, W) \leftarrow \text{shape}(x)$ 
3:  $x_{\text{flat}} \leftarrow \text{reshape}(\text{permute}(x, N, H, W, C), (N \cdot H \cdot W, C))$ 
4: if force_positive then
5:    $W1 \leftarrow \text{ReLU}(W1)$ 
6:    $W2 \leftarrow \text{ReLU}(W2)$ 
7: end if
8:  $D_{\text{mid}} \leftarrow \text{clamp}(\text{ReLU}(D_{\text{mid}}), \text{max} = 0.9 \cdot (t_{\max} - t_{\min}))$ 
9:  $t_{\text{mid}} \leftarrow \text{CallSpiking}(x_{\text{flat}}, W1^T, D_{\text{mid}}, t_{\min}, t_{\max})$ 
10:  $D_{\text{out}} \leftarrow \text{clamp}(\text{ReLU}(D_{\text{out}}), \text{max} = 0.9 \cdot (t_{\max} - t_{\min}))$ 
11:  $t_{\text{blk}} \leftarrow \text{CallSpiking}(t_{\text{mid}}, W2^T, D_{\text{out}}, t_{\min}, t_{\max})$ 
12:  $t_{\text{blk}} \leftarrow \text{permute}(\text{reshape}(t_{\text{blk}}, (N, H, W, C)), N, C, H, W)$ 
13:  $t_{\text{out}} \leftarrow \min(t_j, \text{DropPath}(t_{\text{blk}}))$  // time-domain residual fusion
14: return  $t_{\text{out}}$ 
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Algorithm 2 CallSpiking (Analytic TTFS Spike-Time Mapping)

Input: $t_j \in \mathbb{R}^{B \times C_{\text{in}}}$ (input spike times)
 $W \in \mathbb{R}^{C_{\text{in}} \times C_{\text{out}}}$ (weights), $D \in \mathbb{R}^{C_{\text{out}}}$ (delay),
 $t_{\min}$, $t_{\max}$

Output: $t_i \in \mathbb{R}^{B \times C_{\text{out}}}$ (output spike times)

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1:  $\theta \leftarrow (t_{\max} - t_{\min}) - D$  // threshold term (Eq. 18 form)
2:  $\Delta \leftarrow t_j - t_{\min}$ 
3:  $t_i \leftarrow (\Delta \cdot W) + \theta + t_{\min}$ 
4:  $t_i \leftarrow \text{where}(t_i < t_{\max}, t_i, t_{\max})$  // clamp late/non-spikes to  $t_{\max}$ 
5: return  $t_i$ 
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